

Pratham Arora

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Research Interests

Human-Computer Interaction, Human-AI Interaction, Vision-Language Models, Applied Machine Learning

Education

Plaksha University

B.Tech. in Computer Science & Artificial Intelligence

CGPA: 8.35 / 10.0

July 2026

Mohali, Punjab, India

Research Experience

Undergraduate Researcher, Vision-Language Model Benchmarking

Jan 2026 – May 2026

Supervisor: Prof. Pankaj Pansari, Plaksha University

Mohali, India

- Served as primary data collector for a 459-image benchmark with instrument-verified ground truth across 5 physical estimation tasks, personally performing all instrument measurements and building a 3-tier blur-degradation pipeline (OpenCV) to test model perceptual robustness.
- Independently ran the initial round of experiments across the VLM suite (Gemini 3.1 Pro, Gemini-Robotics-ER, GPT-5.4) before the rest of the team joined, then contributed to experimental design decisions on batch structure and run repetition as the team scaled to cover the remaining models.
- Sole author of the model error analysis: reviewed the 10 worst-performing responses per category per batch across GPT-5.4, Gemini variants, and Gemini-Robotics-ER, identifying a misidentification-vs-miscalibration failure dichotomy: GPT-5.4 fails by misidentifying objects, while Gemini variants correctly identify objects but overestimate weights by $1.8\text{--}2.5\times$; findings directly informed the reasoning-distillation prompts later used to fine-tune open-weight models.
- Co-authored *QUIVER: Benchmarking and Enhancing Physical Reasoning Abilities of Vision-Language Models*, submitted to AAAI 2027 (see *Publications*).

Undergraduate Researcher

Jun 2023 – Aug 2023

Supervisor: Prof. Sandeep Manjanna, Plaksha University

Mohali, India

- Developed a preprocessing pipeline (contrast adjustment, noise reduction, edge detection) to optimize the Segment Anything Model (SAM) for agricultural crop-weed segmentation on sparse datasets, maximizing zero-shot segmentation quality on out-of-distribution agricultural imagery.
- Improved batch processing throughput for 10,000+ image datasets by implementing vectorized NumPy operations and Python multiprocessing.

Publications

- P. Arora**, A. Lodha, Siddharth S. “Wait for It: A Component-Level Analysis of Latency Feedback Mechanisms for LLM Agents in VR.” *Under review, MIT Presence*, 2026.
- V. Lalwani, A. Shafiq, **P. Arora**, M. K. Gurumurthy, P. Pansari. “QUIVER: Benchmarking and Enhancing Physical Reasoning Abilities of Vision-Language Models.” *Submitted to AAAI*, 2027.

Technical Projects

VR LLM Conversational Agent

Aug 2025 – Dec 2025

Unity, Gemini API, Google Cloud STT/TTS

- Engineered a real-time VR conversational agent integrating Gemini 2.5 Flash with Google Cloud STT/TTS, achieving average response latency of 1.8s through parallel API calls, caching, and audio preprocessing.
- Designed a within-subjects experimental study ($N = 18$) evaluating three latency feedback mechanisms (embodied gestures, visual indicators, verbal fillers) using the Godspeed questionnaire, covering dimensions including perceived animacy and immersion.
- Proved via Wilcoxon signed-rank test ($p < 0.05$) that embodied gestures significantly outperform visual loading indicators; co-authored *Wait for It: A Component-Level Analysis of Latency Feedback Mechanisms for LLM Agents in VR*, under review at MIT’s Presence (see *Publications*).

PyTorch, U-Net

- Achieved a Top 100 finish (top 17%) in a DrivenData satellite imagery competition using a U-Net with EfficientNet-B3 backbone for kelp forest segmentation in noisy real-world imagery.
- Built a streak-detection pipeline using an ensemble of CNN classifiers (ResNet, DenseNet) with hard voting to filter atmospherically corrupted training images, achieving an F1 score of 0.952.

Industry Experience

AI & Automation Architect

Jul 2026 – Present

*Practus Advisors**Mumbai, India*

- Building automations for the growth office's reporting workflows to cut manual work and help the team extract more value from existing reports.
- Contributing to the firm's in-house ERP build, focused on finance data migration.

Freelance Web Developer

May 2026 – Present

*PQRS Research (Dr. Niket Tandon)**Remote*

- Leading a full redesign and feature buildout of the PQRS Research website in Next.js, including a dual onboarding system with distinct registration and intake flows for mentors and mentees.

AI Engineering Intern

Jun 2025 – Jul 2025

*Cotality**Kolkata, India*

- Architected a FastAPI backend with AST-based parsing for 15+ languages, enabling automated documentation generation across polyglot codebases at 50,000+ LOC/day.
- Cut AI embedding costs by 85% by building a hash-based system that fingerprints each function's AST and skips re-embedding unchanged code, eliminating redundant Azure OpenAI API calls.
- Built a production RAG pipeline using LangChain and Azure OpenAI, backed by Cosmos DB with IVF vector indexing for semantic code retrieval, serving 17+ enterprise repositories; migrated from FAISS to support persistent, production-scale storage.
- Shipped 12 production REST API endpoints with typed Pydantic schemas, powering real-time documentation preview in the client-facing web interface.

Leadership & Involvement

- **Head of Technology**, Athleda, Plaksha Sports Society (Apr 2025 – Feb 2026): built and maintained a sports platform serving 500+ students, mentoring 2 junior developers through code reviews and production deployments.
- **Web Developer to Head of Web Development**, LEAP.AI @ Plaksha (Feb 2024 – Feb 2025): led a team of 7–8 through 2 full platform redesigns, growing the AI club site to 800+ monthly users.
- **Cohort Representative**, CSAI'22, Plaksha University (Oct 2024 – Jul 2026): coordinated grading-component scheduling across professors to prevent burnout, and built a web tool for students to plan their semester calendar around elective options.

Technical Skills

- **Languages:** Python, SQL, TypeScript
- **ML & Computer Vision:** PyTorch, OpenCV, Segment Anything (SAM), LangChain
- **Systems & Cloud:** FastAPI, Node.js, Next.js, REST APIs, Pydantic, Azure (Cosmos DB, OpenAI), Firebase, FAISS, Google Cloud, Git/GitHub, Vercel